IAT New Test 20

Question paper, answer key and worked solutions

60 questions 180 minutes 240 marks

Biology 1–15; Chemistry 16–30; Mathematics 31–45; Physics 46–60.

Correct +4; incorrect -1; unattempted 0. All 60 questions are available.

This is a provisional 2026-pattern authoring draft for 2027 preparation. Independent academic review is pending.

Instructions

Each item has exactly one correct option. Use the separate solution section for review after attempting the paper. **Figures and equations are part of the questions.**

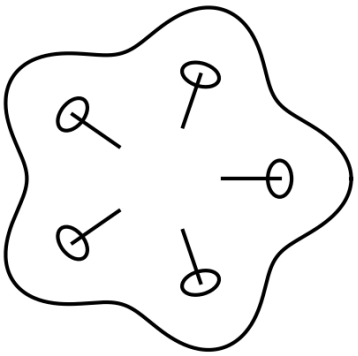
Biology

Q1 Three preparations are examined. P has a cell wall, cytoplasmic ribosomes and division on a suitable cell-free nutrient medium. Q contains a nucleic-acid genome within a protein coat and multiplies only in host cells. R is a small circular infectious RNA without a protein coat. Which assignment is supported by all the observations?

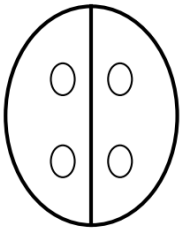
A) P virus; Q moneran; R viroid
B) P moneran; Q virus; R viroid
C) P moneran; Q viroid; R virus
D) P viroid; Q virus; R moneran

Q2 A flowering plant has alternate leaves and the illustrated bisexual, actinomorphic flower. Five stamens are attached to the united corolla. The two fused carpels form a superior ovary with axile placentation; fruit is a berry or capsule. Which family best matches the combined evidence?

Floral organisation: schematic, not to scale



United corolla; five attached stamens



Ovary cross-section

K(5), C(5), A5; superior G(2)

A) Fabaceae
B) Liliaceae
C) Solanaceae
D) Poaceae

Q3 An isolated lysosomal fraction contains a protein-digesting enzyme. An added large protein substrate cannot cross intact lysosomal membranes. Little digestion occurs with intact vesicles in pH 5 buffer; detergent disruption gives strong digestion at pH 5 but little at pH 7. Enzyme amount is unchanged. Which explanation fits both interventions?

A) Membrane sequestration and an acidic activity optimum restrict where digestion occurs
B) The enzyme is absent until detergent synthesises it
C) The substrate is digested only when the lysosomal membrane pumps ATP into it
D) Detergent raises protein concentration equally at both pH values, fully explaining the difference

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Q4 In a stipulated respiratory preparation, one glucose completes glycolysis and both pyruvates are oxidised to acetyl-CoA, but acetyl-CoA cannot enter the Krebs cycle. NAD+ is regenerated by an external system that makes no ATP. Count only net substrate-level ATP and CO2 from those completed stages. What are the two yields per glucose?
A) 4 ATP and 2 CO2
B) 2 ATP and 6 CO2
C) 0 ATP and 0 CO2
D) 2 ATP and 2 CO2

Q5 In a simplified blood-antigen model, a recipient's red cells agglutinate with anti-A and anti-B test sera but not anti-D. The recipient already has anti-D antibodies after a previous sensitisation. Donors are D1=A Rh-negative, D2=O Rh-positive and D3=AB Rh-negative. Ignore all non-ABO/Rh antigens. Which packed-red-cell and plasma pairing avoids the stated antigen-antibody conflicts?
A) D2 red cells and D3 plasma
B) D3 red cells and D1 plasma
C) D1 red cells and D3 plasma
D) D1 red cells and D2 plasma

Q6 In an idealised male reproductive preparation, both vasa deferentia are interrupted while testes, their blood supply and accessory glands remain functional. Which observation distinguishes this intervention from removal of the testes?
A) Semen contains no sperm despite maintained testosterone and ongoing sperm formation within the testes
B) No semen is produced because all seminal fluid originates in the testes
C) Testosterone immediately disappears because it normally travels through the vas deferens
D) The urethra cannot carry urine because the vas deferens is its only outlet

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Q7 Two candidate donors have identical profiles at every VNTR locus assayed in a DNA-fingerprinting experiment. A trace sample matches that shared profile. With no other identifying evidence, which conclusion is justified?
A) The trace must contain equal DNA amounts from both candidates
B) The trace is compatible with either candidate, so these assayed markers do not identify which one contributed it
C) The candidates must differ at every untested genomic position
D) Matching profiles prove that no mutation ever occurred in either candidate

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Q8 Three transmission models are considered: malaria, filariasis and typhoid. An intervention prevents mosquito bites but does not change drinking-water contamination, sanitation or existing infections. Which pair is expected to have reduced new vector-mediated transmission, and why?
A) Malaria and typhoid, because both multiply in mosquito salivary glands
B) Filariasis and typhoid, because both require mosquito entry into the intestine
C) Typhoid alone, because all mosquito-borne infections are bacterial
D) Malaria and filariasis, because both include a mosquito-mediated entry route

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Q9 A recombinant bacterium makes a human proinsulin-like precursor but does not remove its connecting C peptide. Separately produced mature A and B chains can be joined with the correct disulfide bonds to give active insulin. Which interpretation best connects the results?
A) A peptide bond joining the mature A and B chains is the missing permanent insulin linkage
B) Bacteria cannot translate any human-derived amino-acid sequence
C) Precursor processing and correct chain assembly are distinct requirements beyond making the initial polypeptide
D) The C peptide must remain in every mature insulin molecule for receptor binding

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Q10 A two-age expected-population model starts with 40 juveniles and 80 adults. During each interval, half the initial juveniles survive and become adults; three-quarters of initial adults survive; each initial adult produces b new juveniles that are all present at the next census. In interval 1, 10 adults immigrate and 10 adults emigrate after reproduction. The next census totals 200. With the same b and survival rates but no migration in interval 2, what are b and the expected total at its end?
A) b=1.75; total 260
B) b=1.50; total 240
C) b=1.50; total 300
D) b=1.00; total 200

Q11 Island birds trace to one ancestral lineage but have diversified into several beak forms associated with different foods. In a separate comparison, distantly related mammalian lineages independently evolve similar burrowing body forms. Which interpretation uses ancestry as well as function?
A) The birds illustrate adaptive radiation; the similar burrowing forms illustrate convergence
B) The birds illustrate convergence because all eat; the mammals illustrate one shared recent radiation
C) Both observations prove that useful forms arise by directed mutation in each individual
D) Both comparisons demonstrate identical ancestry because ecological functions match

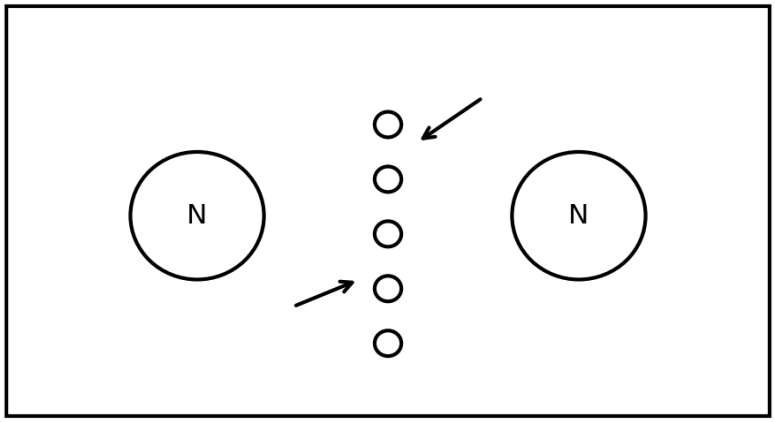
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Q12 A respiratory-volume model has tidal volume 500 mL, inspiratory reserve 2000 mL, expiratory reserve 1000 mL and residual volume 1200 mL. What maximum volume can be expired after a maximal inspiration, without including air that remains after maximal expiration?
A) 4700 mL
B) 3000 mL
C) 3500 mL
D) 1700 mL

Q13 Two hormone signals both promote mobilisation of glucose reserves. P is released rapidly from the adrenal medulla during acute stress. Q is a peptide released from pancreatic islets during fasting. Which assignment is correct?
A) P cortisol; Q insulin from beta cells
B) P adrenaline; Q insulin from alpha cells
C) P thyroxine; Q glucagon from beta cells
D) P adrenaline; Q glucagon from alpha cells

Q14 A plant cell has completed nuclear division. The figure shows Golgi-derived vesicles gathering between the two nuclei. A treatment now prevents those vesicles from fusing, but leaves the completed chromosome segregation intact. Which immediate outcome is most directly predicted?

Plant cell after chromosome segregation



Golgi-derived vesicles; fusion blocked

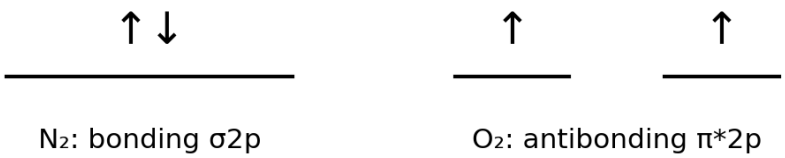
A) Two anucleate cells arise by a cleavage furrow
B) The one cell retains two daughter nuclei and fails to build the initial cell plate
C) The two nuclei fuse automatically into a haploid nucleus
D) DNA replication must reverse before cytokinesis can begin

Q15 In a simplified early mammalian-development experiment, preparation X forms a blastocyst with functional trophoblast but loses its inner cell mass before attachment. Preparation Y retains an inner cell mass but its trophoblast cannot adhere to the endometrium. Other required conditions are maintained. Which paired prediction best separates the two lineage roles?
A) X may attach and still form a normal embryo proper; Y implants because trophoblast adhesion is unnecessary after blastocyst formation
B) X cannot attach solely because its inner cell mass is absent; Y's inner cell mass replaces the trophoblast and restores implantation
C) Neither preparation can have reached a blastocyst because either lineage defect necessarily prevents the earlier cleavage divisions
D) X may begin attachment but cannot form a normal embryo proper; Y can form a blastocyst but cannot implant by the specified route

Chemistry

Q16 The frontier-orbital diagram shows the highest occupied bonding sigma-2p orbital of N2 as paired and the two highest antibonding pi-star orbitals of O2 as singly occupied. All omitted occupied orbitals are paired. One electron is removed from the highest occupied level of each molecule. Which comparison follows?

Frontier orbital occupancies before ionisation



N_2: bonding σ2p O_2: antibonding π*2p

All omitted occupied orbitals are paired

A) Both bonds strengthen and both ions have two unpaired electrons
B) N2 strengthens while O2 weakens; neither ion has an unpaired electron
C) N2 weakens while O2 strengthens; each ion has one unpaired electron
D) Both bonds weaken and the ions remain as magnetically paired as their neutral molecules

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Q17 An iron cyanide complex is either low-spin Fe(II) or low-spin Fe(III), has six CN- ligands and is isolated as a potassium salt. Iron has Z=26 and the measured spin-only moment per complex is sqrt3 Bohr magnetons. With $\mu = \sqrt{u(u+2)}$ for u unpaired electrons, which formula and name are consistent?
A) K3[Fe(CN)6]; potassium hexacyanidoferrate(III)
B) K4[Fe(CN)6]; potassium hexacyanidoferrate(II)
C) K3[Fe(CN)6]; potassium hexacyanidoiron(III)
D) K3[Fe(CN)6]; potassium hexacyanidoferrate(VI)

Q18 Yellow chromate becomes orange dichromate on acidification, without adding a reducing agent. The dichromate is then separately reduced to two Cr3+ ions in acid. Which statement distinguishes these two changes?
A) Both colour changes necessarily represent electron transfer
B) Acidification reduces chromium from +6 to +3 before the reducing agent is added
C) The first change oxidises chromium, while the second requires no electrons
D) The first preserves chromium's +6 state; the second accepts six electrons per dichromate ion

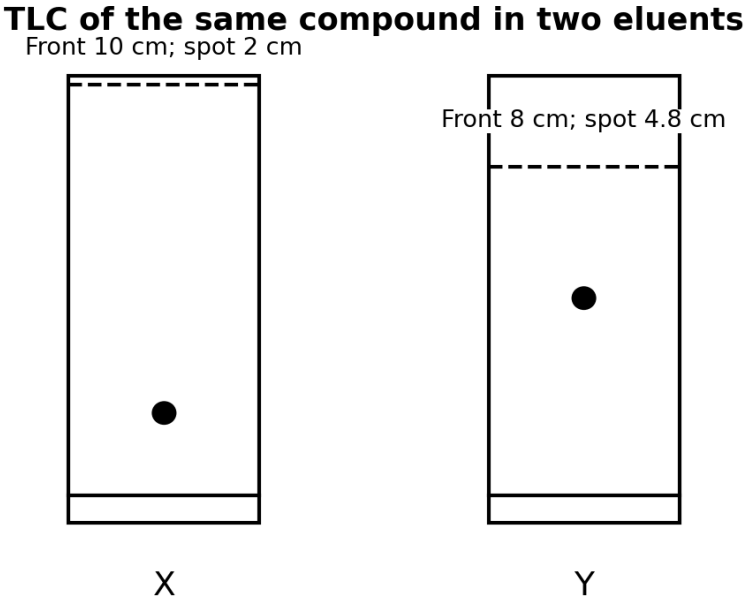
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Q19 A table gives chlorine a covalent radius of 99 pm and argon a van der Waals radius of 188 pm. A student calls the increase a measured covalent-radius jump between adjacent period-3 elements. What is the principal flaw?
A) Chlorine and argon are in different principal shells
B) The two entries use different operational definitions, so they do not directly give a like-for-like covalent-radius trend
C) Argon has fewer protons than chlorine
D) A van der Waals radius is always a negative distance

Q20 Consider only photons from these six downward transitions: in H, 3 to 2, 3 to 1 and 2 to 1; in He+, 6 to 4, 6 to 2 and 4 to 2. Ignore all other transitions and fine structure. With $E_n = -E_0 Z^2/n^2$, how many distinct photon energies occur among the six specified transitions?
A) 6
B) 4
C) 3
D) 2

Q21 A 0.500 g sample of CuSO4.xH2O loses 0.180 g on heating to anhydrous CuSO4. Only water is lost. Take molar masses CuSO4=160 g/mol and H2O=18.0 g/mol. All residue is dissolved to 100.0 mL. What are x and the moles of CuSO4 units in a 25.0 mL aliquot?
A) x=5; 5.00x10^-4 mol
B) x=5; 2.00x10^-3 mol
C) x=4; 5.00x10^-4 mol
D) x=2; 1.00x10^-3 mol

Q22 The same pure compound is run on identical silica TLC plates with two different eluents. Plate X has solvent front at 10.0 cm and spot at 2.0 cm above the baseline. Plate Y has front at 8.0 cm and spot at 4.8 cm. Which interpretation is supported by the figure?



A) Rf changes from 0.20 to 0.40; the migration ratio measures reaction yield
B) Rf changes from 0.80 to 0.60; the compound must decompose in Y
C) Rf is unchanged because a pure compound has one solvent-independent migration constant
D) Rf changes from 0.20 to 0.60; changing eluent can change migration without demonstrating chemical conversion

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Q23 Separate aliquots of but-2-yne undergo (i) controlled hydrogenation over Lindlar catalyst and (ii) acid-catalysed hydration with HgSO4. Which principal products, in the same order, are expected?
A) trans-But-2-ene and butan-2-one
B) cis-But-2-ene and butanal
C) cis-But-2-ene and butan-2-one
D) Butane and butan-2-ol

Q24 P is CH3-CHBr2 and Q is BrCH2-CH2Br. Each undergoes complete double dehydrohalogenation using suitable strong-base conditions, including a base strong enough for the vinyl-halide stage. Which classification and common product are correct?
A) P vicinal; Q geminal; ethene
B) P geminal; Q vicinal; ethyne
C) P geminal; Q vicinal; ethane
D) P vicinal; Q geminal; bromoethene

Q25 At room temperature, 2-methylpropan-2-ol gives rapid turbidity in the Lucas test, while phenol does not give the corresponding alkyl-chloride turbidity. Why does the negative phenol result not identify it as a primary alcohol?
A) All negative Lucas results prove the complete absence of an OH group
B) Phenol forms the same tertiary carbocation but its aromatic ring hides the precipitate
C) A phenolic OH is necessarily a primary aliphatic carbinol group
D) Lucas classification applies to aliphatic alcohols; phenolic OH is bonded to aromatic sp2 carbon

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Q26 A single linear tetrapeptide has sequence Cys-Gly-Ala-Cys. Its two cysteine side chains are joined by an intramolecular disulfide bond. A selective treatment hydrolyses every peptide bond but leaves the S-S bond intact. Per original peptide molecule, how many water molecules are consumed in peptide hydrolysis and how many separate covalent product molecules remain?
A) 3 water; 3 product molecules
B) 3 water; 4 product molecules
C) 4 water; 3 product molecules
D) 2 water; 3 product molecules

Q27 Ideal buffers X and Y each occupy 1.00 L. X contains 0.0100 mol weak base B and 0.0100 mol BH+; Y contains 0.100 mol of each. The base has pKb=5.00 at 25 C. Add 0.0200 mol HCl to each separately, with negligible volume change. Using the appropriate dominant equilibria, what are the approximate final pH values of X and Y? Logs are base 10.
A) 2.00; $9 + \log(3/2)$
B) 9.00; 9.00
C) 2.00; $9 + \log(2/3)$
D) 1.70; $9 + \log(2/3)$

Q28 A binary liquid feed has mole fraction 0.40 of A. In a single equilibrium partial-vaporisation step, the resulting liquid has A mole fraction 0.30 and the resulting vapour has A mole fraction 0.60. No material is lost. What fraction of total feed moles is in the vapour, and which phase is enriched in A relative to the feed?
A) 2/3; vapour
B) 1/2; liquid
C) 1/3; liquid
D) 1/3; vapour

Q29 A cell reaction is $\mathrm{A(s)} + \mathrm{B^{2+}(aq)} \longrightarrow \mathrm{A^{2+}(aq)} + \mathrm{B(s)}$, with E0=0.3000 V at 298 K. Free A2+ activity is 0.0100; complexation leaves free B2+ activity 10^-6 although its total analytical concentration is unchanged. Only free B2+ is reduced. A report doubles every coefficient of the cell equation. Using $E = E^0 - (0.0592/n)\log Q$, what potential does the doubled equation predict and what is its actual Gibbs-change ratio to the original equation at this composition?
A) 0.1816 V; ratio 2
B) 0.2408 V; ratio 2
C) 0.1816 V; ratio 1
D) 0.0632 V; ratio 2

Q30 At a stated temperature, $\mathrm{H_2(g)} + \frac{1}{2}\,\mathrm{O_2(g)} \longrightarrow \mathrm{H_2O(l)}$ has enthalpy change -286.0 kJ/mol, and vaporising one mole of water has enthalpy change +44.0 kJ/mol. Take RT=2.50 kJ/mol and ideal reactant/product gases. What are delta H and delta U for forming one mole of H2O(g) from those elements?
A) -286.0 and -282.25 kJ/mol
B) -242.0 and -240.75 kJ/mol
C) -242.0 and -243.25 kJ/mol
D) -242.0 and -238.25 kJ/mol

Mathematics

Q31 Let $A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$ and $B_t = \begin{pmatrix} t^2 & t & 1 \\ 1 & 2t+3 & t \\ t & 1 & 15-2t \end{pmatrix}$, where t is real. Among members of this family that commute with A, what is det Bt?
A) 676
B) 756
C) 0
D) 728

Q32 Five distinct people A,B,C,D,E occupy positions 1 through 5, one each. A cannot occupy 1 or 2, B cannot occupy 2 or 3, and C cannot occupy 3 or 4. How many arrangements satisfy all restrictions?
A) 36
B) 28
C) 24
D) 20

Q33 Let A have six labelled elements and B have four labelled elements. For a map f:A→B define xRy when f(x)=f(y). How many equivalence relations on A have class sizes 3,2,1, and for each such fixed relation how many maps f induce exactly that relation?
A) 60 and 64
B) 10 and 24
C) 60 and 24
D) 60 and 4

Q34 For real a,b, the polynomial $z^2 - 2(a+i)z + b + 6i$ has a repeated complex root. What is a+b?
A) 8
B) 10
C) 12
D) 11

Q35 A differentiable function F on the real line satisfies $F'(x) = (x+1)/(x^2 - 2x + 2)$ and F(0)=0. What is F(2)?
A) π
B) π/2
C) ln 2
D) 2π

Q36 A line meets the planes x=0 and y=0 at A and B, respectively. P=(2,3,4) divides AB internally with AP:PB=2:1. The line is parallel to the plane x+y+z=0. Which pair gives B and the length AB?
A) (6,0,0); 3√14
B) (3,0,6); 3√14
C) (3,0,0); 3√10
D) (3,0,6); √14

Q37 For the real parametric curve $x = t^2,\ y = t^3 - t$, classify the tangent at the origin and find the angle between the two distinct tangents at the self-intersection (1,0).
A) Regular vertical tangent at the origin; right angle at (1,0)
B) Singular point at the origin; right angle at (1,0)
C) Regular horizontal tangent at the origin; angle π/4 at (1,0)
D) Regular vertical tangent at the origin; coincident tangents at (1,0)

Q38 The data set {0,0,0,4} is changed to {0,0,0,-4}. Using population variance and mean absolute deviation about the arithmetic mean, which statement is correct?
A) Both measures change
B) Variance stays 3; mean absolute deviation stays 3/2
C) Variance stays 3; mean absolute deviation changes to 2
D) Variance becomes 4; mean absolute deviation stays 1

Q39 For real a,b the limit $\lim_{x\to 0}(\sin x + a\cos x + b - x)/x^2$ is finite and equals 3. What is a-b?
A) -6
B) 6
C) -12
D) 3

Q40 Three players play each other once with no draws; the three match outcomes are independent. P(A beats B)=1/2, P(B beats C)=2/3 and P(C beats A)=3/4. What is the probability that the results are transitive, meaning that one player beats both others?
A) 1/3
B) 1/4
C) 3/4
D) 17/24

Q41 For x>0 let $S(x) = \arctan(2x) + \arctan x$, with principal arctangent values. Which expression is correct for every x>1/√2?
A) $\arctan(3x/(1-2x^2))$
B) $\pi + \arctan(3x/(1-2x^2))$
C) $\arctan(3x/(1-2x^2)) - \pi$
D) $\pi/2 + \arctan(3x/(1-2x^2))$

Q42 For y>0 solve $dx/dy + x/y = y$ with x=2 at y=1. What is x when y=2?
A) 7/2
B) 8/3
C) 3
D) 13/6

Q43 The distinct polynomials $P(x) = x^2 + px + q$ and $Q(x) = x^2 + qx + p$ have real coefficients and exactly one distinct common root. That root is a repeated root of P. What is p+q?
A) 1
B) -3
C) 0
D) -1

Q44 A line joins two distinct intersection points of the hyperbola $x^2/4 - y^2 = 1$ and the circle $x^2 + y^2 = 9$. Only lines parallel to neither coordinate axis are allowed. How many such lines exist, and what is the length of each corresponding chord?
A) Two lines; length 6
B) Four lines; length 6
C) Two lines; length 3
D) Two lines; length 4√2

Q45 In the oriented xy-plane let a=(2,1) and J(a)=(-1,2), its counterclockwise quarter-turn. A vector v satisfies v·a=10 and the signed determinant det(a,v)=15. What is |v|?
A) 5
B) √13
C) √65
D) √85

Physics

Q46 A proposed drag law is $F = K\rho^a v^b L^c$, where ρ is density, v is speed, L is length and K is dimensionless. Which result follows from dimensional analysis alone?
A) a=1,b=2,c=2; K is not fixed
B) a=1,b=1,c=2; K=1
C) a=1,b=2,c=1; K is not fixed
D) a=1,b=2,c=2; K must be 1

Q47 On a smooth horizontal line, a 1 kg block A is immediately left of a 2 kg block B. They just touch and are initially at rest. External forces 3 N left on A and 12 N right on B are switched on. The contact cannot transmit tension. What are their initial accelerations, taking right positive?
A) Both +3 m s−2
B) A=-3 m s−2, B=+6 m s−2
C) A=+3 m s−2, B=-6 m s−2
D) Both zero

Q48 A guided rigid platform moves vertically without tilting and is supported by two initially unstretched parallel rods. Their elastic stiffnesses are k and 2k. Rod 1 stays elastic up to 100 N. Rod 2 reaches 150 N and then is ideally plastic, maintaining that force while extending. Loading is quasistatic. At total load 240 N, what are the rod forces and the extension divided by its value when rod 2 first yielded?
A) 80 N,160 N; 16/15
B) 100 N,140 N; 4/3
C) 90 N,150 N; 6/5
D) 120 N,120 N; 8/5

Q49 A body obeying Newton’s law of cooling in a constant ambient temperature is observed at 80,65 and 55 degrees C at three equally spaced times. What is its predicted temperature after one further equal interval?
A) 45 degrees C
B) 50 degrees C
C) 40 degrees C
D) 145/3 degrees C

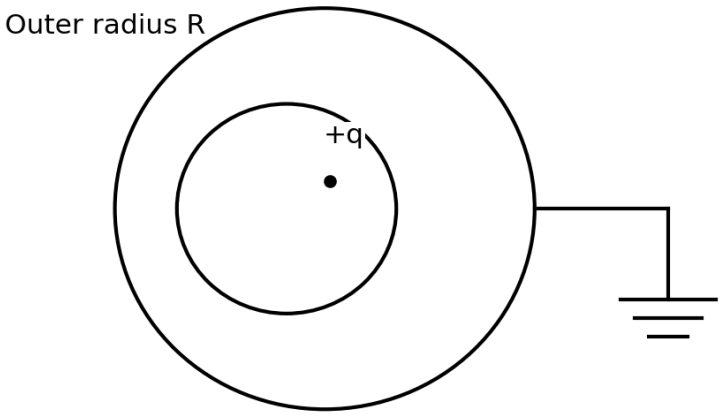
Q50 One mole of a monatomic ideal gas at 100 K mixes with three moles of the same gas at 300 K in an isolated rigid vessel. What is the final rms speed divided by the initial rms speed of the hotter gas?
A) √(5/6)
B) √(2/3)
C) 5/6
D) √3

Q51 A mass m on a horizontal spring k starts at rest at its natural equilibrium. A constant force F is switched on for time π/(2ω), where ω=√(k/m), and then removed instantly without an impulse. What is the amplitude of the subsequent free oscillation about the original equilibrium?
A) F/k
B) √2 F/k
C) 2F/k
D) F/(√2 k)

Q52 An ideal string of length 1.00 m has a fixed end at x=0 and a free end at x=1.00 m. In a standing-wave mode its nodes are x=0,0.40,0.80 m and the free end is an antinode. Wave speed is 80 m s−1. What are the frequency and its harmonic number relative to the fundamental?
A) 200 Hz; fifth
B) 100 Hz; fourth
C) 100 Hz; fifth
D) 50 Hz; third

Q53 A conductor has a spherical outer surface of radius R and an off-centre point charge +q inside a closed empty cavity, without electrical contact. It is first grounded, then disconnected from Earth, and finally an additional charge +2q is deposited on its outer surface. No external charges are present. Which final description is correct?

Cavity charge and grounding sequence

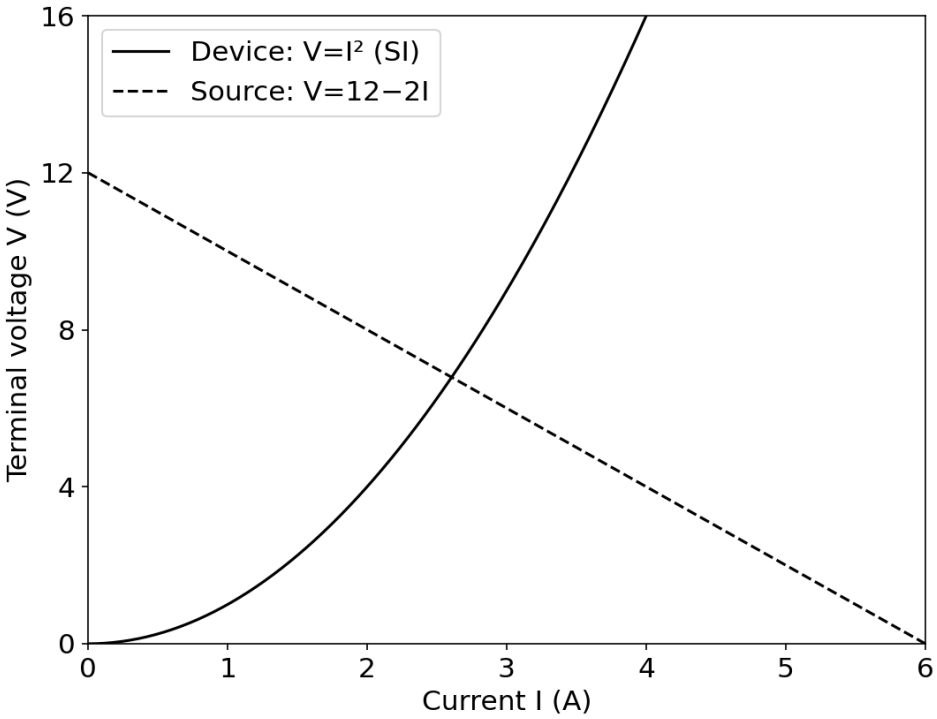


Ground → disconnect → deposit +2q

A) Inner -q, outer +3q; exterior field $3kq/r^2$
B) Inner 0, outer +2q; exterior field $2kq/r^2$
C) Inner -q, outer +2q; exterior field generally lacks spherical symmetry
D) Inner -q nonuniform, outer +2q uniform; exterior field $2kq/r^2$ radially outward

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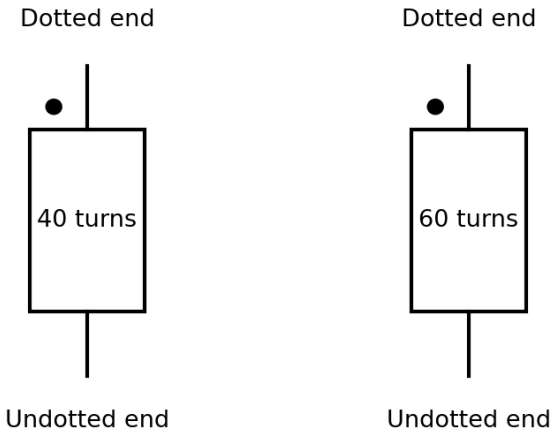
Q54 The graph shows a device characteristic $V=(1\ \mathrm{V\ A^{-2}})I^2$ for $I\geq 0$ and a source load line $V=12\ \mathrm{V}-(2\ \Omega)I$. With no other series element, what is the physical operating current?



A) (√13-1) A
B) 3 A
C) 6 A
D) (√13+1) A

Q55 An ideal transformer has a 100-turn primary supplied with 10 V rms and two isolated secondary windings of 40 and 60 turns. The dot marks in the diagram denote ends with the same instantaneous polarity. In connection X, an undotted end of one secondary joins the dotted end of the other; in Y, the two dotted ends join. What are the open-circuit rms voltages between the free ends in X and Y?

Secondary dot polarities and connection rules

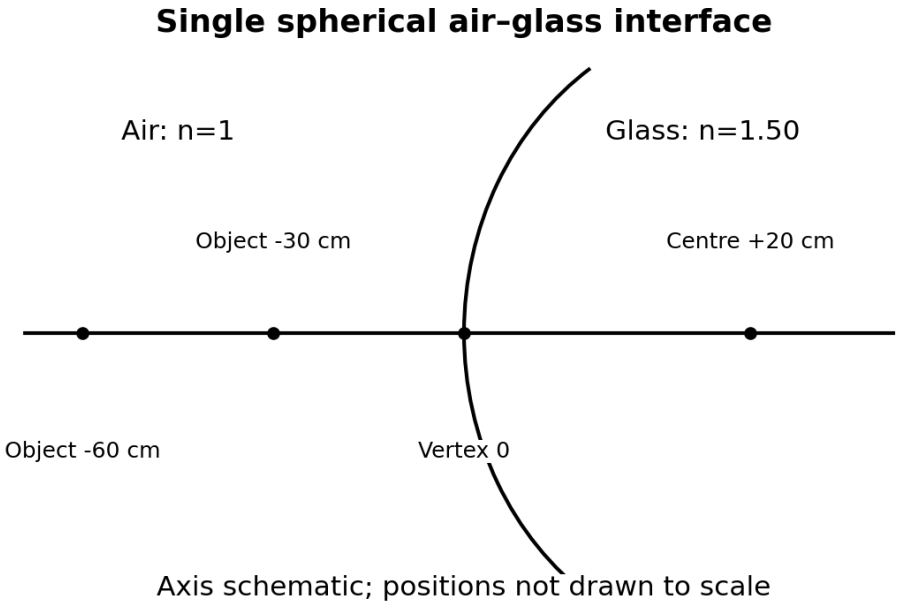


X joins unlike ends; Y joins dotted ends

A) 2 V and 10 V
B) 10 V and 2 V
C) 10 V and 10 V
D) 4 V and 6 V

Q56 A series RLC circuit has R=3 ohm, XL=8 ohm and XC=4 ohm at the operating frequency. The source is 10 V rms. What is the inductor rms voltage, and why can it exceed the source voltage?
A) 8 V; all voltages add as positive scalars
B) 16 V; the circuit creates net energy
C) 16 V; reactive voltages partially cancel as phasors
D) 10 V; no component can exceed the source

Q57 A single spherical refracting surface separates air on the left from glass of index 1.50 on the right. Its vertex is x=0 and centre of curvature x=+20 cm. For axial objects at x=-30 cm and x=-60 cm, respectively, what are the paraxial image coordinates? Use the Cartesian sign convention.



A) +180 cm and +180 cm
B) -180 cm and -180 cm
C) -30 cm and +60 cm
D) -180 cm and +180 cm

Q58 In a Young experiment, slit separation is d, screen distance D, and a point source initially on the axis is distance L before the slits. It is shifted upward by $s=L\lambda/(2d)$. Assume $L,D \gg d,s$ and paraxial paths; upward screen coordinate is positive. What happens at the old screen centre, and where is the new zero-path-difference fringe? Let $\beta=D\lambda/d$.
A) Bright; $+\beta/2$
B) Dark; $+\beta/2$
C) Dark; $-\beta/2$
D) Bright; $-\beta$

Q59 A nonrelativistic electron starts at rest, gains kinetic energy through a 400 V accelerating potential, and then crosses a 100 V retarding potential without reflection. What is its final de Broglie wavelength divided by its wavelength immediately after acceleration?
A) $2/\sqrt{3}$
B) 1/2
C) $\sqrt{3}/2$
D) 4/3

Q60 Two parallel branches contain ideal diodes in opposite directions. The left-to-right conducting branch has a 100 ohm resistor; the right-to-left conducting branch has a 200 ohm resistor. A 6 V battery is applied across the pair, first left positive and then right positive. What are the magnitudes of the two steady battery currents?
A) 30 mA,60 mA
B) 60 mA,30 mA
C) 90 mA,90 mA
D) 0,0

Answer key and worked solutions

Every solution explains the selected choice and the alternative errors.

Biology

Q1 / B Cellular division and capsid evidence separate a moneran, virus and viroid

Answer: (B) P moneran, Q virus and R viroid.

Why: P has cellular machinery and can divide without a living host, which distinguishes it from the acellular infectious preparations. A nucleic-acid genome enclosed by a protein coat is consistent with a virus in Q. The infectious naked circular RNA in R matches a viroid, whose organisation lacks a viral protein capsid. Host dependence alone would not distinguish Q from R; coat and cellular evidence must be used together.

Why not the others: (A) reverses the cellular and coated acellular preparations. (C) assigns a protein coat to a viroid and removes that defining distinction from the virus. (D) calls a dividing ribosome-containing cell a viroid and an acellular RNA a moneran.

Key point: Combine cellular organisation, independent division and the presence of a capsid when classifying an infectious preparation.

Q2 / C A family assignment combines epipetalous stamens, carpel fusion and fruit type

Answer: (C) Solanaceae.

Why: The combination of a five-parted united calyx and corolla, five epipetalous stamens, and a bicarpellary superior ovary with axile placentation is characteristic of Solanaceae in the standard family description. The berry-or-capsule fruit supports that assignment. A single trait such as a superior ovary is shared by several families, so the linked whorl arrangement and ovule attachment are more informative than that trait alone.

Why not the others: (A) ordinarily has the distinctive papilionaceous organisation, ten stamens and a monocarpellary legume-producing gynoecium. (B) has a trimerous floral organisation and usually six stamens rather than the illustrated pentamerous one. (D) has the reduced grass-flower organisation rather than the stated five-petalled epipetalous flower.

Key point: Identify a family from a correlated suite of floral and fruit characters rather than one shared feature.

Q3 / A A substrate-access intervention is distinguished from a lysosomal pH requirement

Answer: (A) Compartmentalisation and acidic conditions restrict digestion.

Why: Before disruption, the externally added substrate is physically separated from the enzyme, even though the surrounding buffer has a favourable pH. Detergent removes that access barrier, revealing substantial activity at pH 5. The weak rate at pH 7 despite equal enzyme amount shows that access alone is insufficient; catalytic groups and structure also depend on an appropriate acidic environment. The two interventions therefore expose different constraints on the same preparation.

Why not the others: (B) contradicts the stated unchanged enzyme amount and gives detergent an unsupported synthetic role. (C) replaces enzyme-substrate access with an unstated ATP-driven mechanism. (D) cannot explain why the disrupted preparations differ when enzyme amount and substrate supply are controlled.

Key point: Separate compartment access from chemical conditions when interpreting an organelle-enzyme assay.

Q4 / D Respiratory-stage accounting stops at acetyl-CoA despite cofactor recycling

Answer: (D) 2 ATP and 2 CO2.

Given: Glycolysis and two pyruvate-oxidation reactions finish; no Krebs turnover or external ATP formation occurs.

Formula/Principle: Add the stage-specific carbon and substrate-level ATP balances rather than the full aerobic yield.

Step 1: Glycolysis converts one six-carbon glucose into two three-carbon pyruvates, releases no CO2 and has net ATP yield

$$4 - 2 = 2 \text{ ATP}.$$

Step 2: Each pyruvate loses one carbon as CO2 on conversion to a two-carbon acetyl group. Thus

$$n(\text{CO}_2) = 2(1) = 2 \text{ molecules per glucose}.$$

Step 3: Krebs-cycle ATP and CO2 are absent because its stated entry step is blocked.

Why: Producing acetyl-CoA is not the same as oxidising its two carbons through the cycle. External NAD+ regeneration allows the named earlier reactions to continue without silently adding oxidative-phosphorylation ATP. This is a defined stage-accounting model, not the total yield of an intact cell oxidising glucose completely.

Why not the others: (A) counts the two glycolytic ATP investments as retained ATP. (B) assigns all six glucose carbons to CO2 despite the retained acetyl groups. (C) assumes the downstream block stops completed upstream reactions even with the supplied cofactor regeneration.

Key point: Reconstruct ATP and carbon outputs only from stages that actually operate.

Q5 / C An antigen assay and sensitisation history constrain two different blood components

Answer: (C) D1 red cells with D3 plasma.

Why: Reaction with both ABO antisera identifies A and B antigens on the recipient's cells, while the negative anti-D test identifies absence of the D antigen. The pre-existing anti-D makes D-positive donor red cells incompatible in this stated model. D1's A-positive, D-negative red cells introduce neither an ABO antigen foreign to this AB recipient nor D. D3's AB plasma lacks the ordinary anti-A and anti-B antibodies that would otherwise attack the recipient's cells. Red-cell and plasma compatibility must be analysed in opposite directions because one supplies antigens and the other supplies antibodies.

Why not the others: (A) introduces D-positive cells into a recipient with anti-D. (B) supplies group-A plasma containing anti-B against the recipient's B-bearing cells. (D) supplies group-O plasma with anti-A and anti-B against the recipient's AB cells.

Key point: In an antigen-antibody model, analyse donor-cell antigens and donor-plasma antibodies separately; actual transfusion practice additionally requires clinical compatibility procedures.

Q6 / A A reproductive-duct intervention separates transport, gametogenesis and endocrine function

Answer: (A) Sperm transport is interrupted while testicular functions persist.

Why: The vas deferens carries sperm away from the testis-associated storage and transport pathway. Interrupting that route prevents sperm from reaching the ejaculate, but does not remove the tissue that produces sperm or the Leydig cells that release testosterone into blood. Accessory glands can still contribute seminal secretions. The observation therefore separates transport failure from loss of endocrine and gamete-producing organs.

Why not the others: (B) ignores accessory-gland contributions to seminal fluid. (C) assigns a blood-borne hormone to the sperm duct as its necessary transport route. (D) confuses the reproductive duct connection with the separate bladder-to-urethra urinary route.

Key point: A blocked reproductive transport route need not abolish gamete formation or endocrine secretion.

Q7 / B Identical assayed VNTR profiles limit individual identification without implying a mixture

Answer: (B) Either candidate is compatible with the assayed profile.

Why: A profile is a set of observations at selected variable loci. When both candidates share those observations, matching the trace does not distinguish them within the assay's information. It also does not demonstrate that both contributed DNA, because one donor alone could produce the same result. Broader identity or mutation-history claims require evidence beyond the markers actually measured.

Why not the others: (A) turns compatibility with two sources into proof of a two-source mixture and an unmeasured mixture ratio. (C) infers universal difference at unobserved positions from equality at the observed ones. (D) makes an absolute historical claim that a finite marker panel cannot establish.

Key point: State what a fingerprint can distinguish in the candidate set, not what its name suggests it should identify.

Q8 / D A vector-exposure control distinguishes protozoan and helminth routes from typhoid ingestion

Answer: (D) Malaria and filariasis.

Why: Malaria and filarial infection are transmitted through mosquito vectors in the standard disease cycles, even though their causative organisms and vector associations differ. Blocking bites interrupts that shared exposure route without directly killing pathogens already inside a host. Typhoid is associated with ingestion of contaminated food or water rather than a required mosquito-bite stage, so the stipulated intervention does not directly remove its route. This prediction concerns new transmission, not an immediate cure or a measured effect size.

Why not the others: (A) wrongly inserts typhoid into a mosquito cycle. (B) again makes typhoid vector-borne and misplaces the mosquito's role. (C) misclassifies the protozoan and helminth infections as bacteria and reverses the route-based prediction.

Key point: Compare the specific life-cycle entry route when predicting an intervention's direct effect.

Q9 / C Proinsulin processing distinguished from expression and disulfide assembly of mature insulin

Answer: (C) Processing and correct assembly matter beyond translation.

Why: Proinsulin contains a connecting segment that is removed during maturation. Mature insulin retains A and B chains linked by the appropriate disulfide arrangement rather than by retaining that C-peptide connector. Producing the precursor therefore does not guarantee producing the mature active molecule. Activity after correct assembly of separately produced chains shows that expression of human-derived chains is possible, while processing and folding remain separate biochemical tasks.

Why not the others: (A) substitutes a permanent interchain peptide connection for the stated mature disulfide linkage. (B) contradicts production and successful use of the recombinant chains. (D) reverses the role of the connecting peptide during normal maturation.

Key point: Distinguish expression, precursor processing and final protein assembly in a recombinant product.

Q10 / B An observed census identifies reproduction before an age-structured second interval

Answer: (B) b=1.50 and expected total 240.

Given: Only adults present at an interval's start reproduce; survival and maturation precede the next age census.

Formula/Principle: Keep births, surviving cohorts and migration separate before updating the age structure.

Step 1: Adults at census 1 are

$$A_1 = 0.5(40) + 0.75(80) + 10 - 10 = 80.$$

Step 2: New juveniles are $J_1 = 80b$, so

$$80 + 80b = 200 \Rightarrow b = 1.50, \quad J_1 = 120.$$

Step 3: For the second interval,

$$J_2 = 1.5(80) = 120, \quad A_2 = 0.5(120) + 0.75(80) = 120.$$

Step 4:

$$N_2 = J_2 + A_2 = 240.$$

Why: The total count alone is insufficient for the next projection because juveniles and adults have different roles and survival. Births in one interval enlarge the next juvenile cohort but do not become same-interval breeders. Cancelling net migration does not justify omitting maturation or adult mortality. These are stipulated expected counts, not an assumption of logistic growth or a field estimate of carrying capacity.

Why not the others: (A) omits the 20 maturing juveniles when inferring b. (C) uses the 120 juveniles as if they were the current breeding adults in interval 2. (D) neither reproduces the observed census with the stated surviving adults nor updates its age composition.

Key point: Preserve cohort identity when moving from a population balance to a second projection.

Q11 / A Common ancestry with ecological divergence contrasted with independent functional resemblance

Answer: (A) Adaptive radiation in the birds and convergence in the mammalian comparison.

Why: Diversification from one ancestral stock into different ecological roles is the defining pattern of adaptive radiation. Similar adaptive forms appearing independently in distant lineages instead support convergent evolution. Functional similarity alone cannot establish a recent common origin, and different functions can arise among close relatives. The supplied ancestry observations therefore determine how the morphological similarities and differences are interpreted.

Why not the others: (B) discards the one-lineage origin of the birds and invents a shared recent origin for the distant mammals. (C) makes an unsupported claim about directed within-individual mutation rather than population evolution. (D) equates matching function with identical ancestry.

Key point: Use the ancestry pattern to distinguish divergence into niches from independent resemblance.

Q12 / C Vital capacity excludes the residual compartment despite a maximal inspiration

Answer: (C) 3500 mL.

Given: TV=500 mL, IRV=2000 mL, ERV=1000 mL and RV=1200 mL.

Formula/Principle: Vital capacity is the volume between maximal inspiration and maximal expiration.

Step 1:

$$VC = IRV + TV + ERV.$$

Step 2:

$$VC = 2000 + 500 + 1000 = 3500 \text{ mL}.$$

Why: Residual volume stays in the lungs after the specified maximal expiration and is outside the requested moving volume. It contributes to total lung capacity, not to vital capacity. The calculation identifies the two endpoint states before combining the named volumes.

Why not the others: (A) includes residual volume and gives total lung capacity. (B) leaves out ordinary tidal volume. (D) adds residual volume to tidal volume while ignoring both reserves.

Key point: Define the starting and ending respiratory states before adding volume components.

Q13 / D An acute medullary signal and a fasting islet signal share an endpoint but not a source

Answer: (D) Adrenaline and alpha-cell glucagon.

Why: The adrenal medulla supplies catecholamines such as adrenaline in the acute stress response. Pancreatic alpha cells release glucagon, which helps raise available glucose during fasting, whereas beta-cell insulin generally promotes glucose uptake and storage. Similar effects on a measured glucose variable do not make the hormones identical or give them the same tissue source. Source, cell type and physiological trigger jointly identify the pair.

Why not the others: (A) assigns the medullary rapid signal to a cortical hormone and pairs fasting mobilisation with insulin. (B) retains adrenaline but gives insulin the wrong islet cell and action. (C) substitutes thyroid hormone for the medullary signal and places glucagon in beta cells.

Key point: Combine endocrine source, trigger and action rather than identifying a hormone from one endpoint alone.

Q14 / B Vesicle-fusion blockade interrupts plant cytokinesis after successful nuclear separation

Answer: (B) A binucleate cell without the initial cell plate.

Why: In the depicted plant-cell cytokinesis route, vesicles fuse centrally to form the cell plate and associated new partition. Blocking their fusion prevents this cytoplasmic separation even though karyokinesis has already produced two daughter nuclei. No reversal of chromosome segregation is required for the cell to remain binucleate. The observed vesicle location identifies the relevant plant mechanism rather than an animal-type constricting furrow.

Why not the others: (A) substitutes a different cytokinesis mechanism and removes nuclei that the stem says are present. (C) invents nuclear fusion and a ploidy reduction with no stated cause. (D) assigns a backward DNA-synthesis requirement to a membrane-partitioning step.

Key point: Nuclear division and cytoplasmic partitioning are separable events with different cellular machinery.

Q15 / D Complementary blastocyst-lineage defects separate fertilisation, implantation and embryo-proper development

Answer: (D) X separates attachment capacity from embryo formation; Y separates blastocyst formation from implantation.

Why: The trophoblast participates in attachment to the uterine lining and later maternal-fetal interactions, while the inner cell mass supplies the embryo proper. Losing the inner cell mass therefore does not logically erase an already formed trophoblast's initial adhesion capability, but it prevents normal embryo-proper development. Conversely, having an inner cell mass does not substitute for the specified trophoblast adhesion function. The stage chronology also matters: fertilisation and cleavage precede blastocyst attachment, so an implantation defect does not prove those earlier events never occurred. This deliberately limited model does not assert that a trophoblast alone forms every tissue of a complete placenta.

Why not the others: (A) ignores loss of the embryo-forming lineage and the specified adhesion defect. (B) confuses the attachment role of trophoblast with the embryo-proper role of inner cell mass. (C) infers failure of an earlier completed stage from a later functional defect.

Key point: Distinguish stage chronology from lineage-specific functions.

Chemistry

Q16 / C Opposite bonding effects of electron removal give equal final bond orders but new spin counts

Answer: (C) N2 weakens, O2 strengthens, and each ion has one unpaired electron.

Given: Removal occurs from a bonding orbital in N2 and an antibonding orbital in O2.

Formula/Principle: Bond order is $(N_b - N_a)/2$; removing a bonding electron lowers it, while removing an antibonding electron raises it.

Step 1: Neutral N2 has bond order 3. Its ion has

$$BO(\mathsf{N_2}^+) = 3 - 1/2 = 2.5.$$

Step 2: Neutral O2 has bond order 2. Its ion has

$$BO(\mathsf{O_2}^+) = 2 + 1/2 = 2.5.$$

Step 3: The formerly paired N2 frontier orbital becomes singly occupied. Removing one of O2's two frontier electrons leaves one unpaired electron.

Why: Ionisation need not weaken every molecular bond; its effect depends on whether the removed electron supported or opposed bonding. Equal final bond orders do not mean equal changes from the neutral species. Magnetic behaviour must be counted from occupancy rather than inferred solely from the fact that an electron was removed.

Why not the others: (A) treats both removed electrons as antibonding and miscounts the remaining spins. (B) reverses the bonding roles and pairs odd-electron ions. (D) treats both removed electrons as bonding and ignores the changed frontier occupations.

Key point: Identify the removed orbital's bonding character before predicting ionisation effects.

Q17 / A A spin measurement determines oxidation state before salt stoichiometry and anion naming

Answer: (A) K3[Fe(CN)6], potassium hexacyanidoferrate(III).

Given: The allowed oxidation states are +2 and +3; both octahedral candidates are low spin.

Formula/Principle: Use the magnetic moment to infer unpaired count, low-spin d occupancy to identify oxidation state, then balance ligand charge and apply anionic-complex naming.

Step 1:

$$u(u+2) = 3 \Rightarrow u = 1$$

for nonnegative integer u.

Step 2: Fe(II) is d6 and low-spin t_{2g}^6, with zero unpaired electrons. Fe(III) is d5 and low-spin t_{2g}^5, with one, so the metal is Fe(III).

Step 3:

$$q_{\text{complex}} = +3 + 6(-1) = -3,$$

requiring three potassium ions.

Step 4: Six cyanido ligands give hexacyanido; an anionic iron complex takes ferrate, followed by oxidation number III.

Why: A magnetic measurement supplies electron-configuration evidence rather than the oxidation number directly. Low-spin information makes the two permitted oxidation states distinguishable, and the resulting charge determines the salt stoichiometry. The ligand count is a coordination count, not the metal oxidation number.

Why not the others: (B) describes the diamagnetic low-spin Fe(II) candidate. (C) omits the anionic metal-name form. (D) substitutes the ligand count six for the inferred oxidation state three.

Key point: Connect magnetic occupancy, ionic charge and nomenclature as separate deductions.

Q18 / D An acid-driven colour change is separated from a subsequent chromium reduction

Answer: (D) The first preserves +6; the second accepts six electrons.

Given: CrO_4^{2-} interconverts with $Cr_2O_7^{2-}$, then Cr(III) forms.

Formula/Principle: Determine chromium oxidation states from oxygen and ion charge before deciding whether a colour change is redox.

Step 1: In chromate,

$$x + 4(-2) = -2 \Rightarrow x = +6.$$

Step 2: In dichromate,

$$2x + 7(-2) = -2 \Rightarrow x = +6.$$

Step 3: Reduction of both chromium atoms from +6 to +3 accepts

$$2(6-3) = 6 \text{ electrons.}$$

Why: Acidification changes the equilibrium distribution of chromium(VI) oxoanions and their colour, but it does not change the chromium oxidation number. The later reduction does change that number and requires electron uptake. Visible colour alone therefore cannot classify a reaction as oxidation or reduction.

Why not the others: (A) equates a spectral change with redox regardless of oxidation states. (B) assigns the later reduced state to the acid-base interconversion. (C) invents electron loss in the first step and omits electron gain in the second.

Key point: Track oxidation numbers independently of colour when identifying a redox change.

Q19 / B Different radius definitions prevent an unsupported cross-definition periodic trend

Answer: (B) Different operational definitions are being compared.

Why: A covalent radius is associated with bonded-atom separation, whereas a van der Waals radius is obtained from nonbonded contacts. Noble-gas entries commonly use the latter because ordinary covalent-bond data are not comparable to the bonded-atom data used for many other elements. The numerical difference is real for those definitions, but it cannot by itself establish the stated change in one common radius measure or its cause.

Why not the others: (A) puts adjacent period-3 atoms in different principal shells. (C) reverses their proton-number order. (D) denies the meaning of a positive nonbonded-contact distance rather than identifying the definition mismatch.

Key point: Check how a periodic-property measurement is defined before comparing its numbers.

Q20 / C Specified hydrogen-like transitions coincide after simultaneous level and charge scaling

Answer: (C) Three distinct photon energies.

Given: H has Z=1 and He+ has Z=2; every selected He+ level is twice the corresponding H level.

Formula/Principle: A transition energy is the difference between its two level energies, and coincident differences count only once.

Step 1: For a He+ level 2n,

$$E_{2n} = -E_0 \frac{2^2}{(2n)^2} = -E_0/n^2,$$

the same energy as H level n.

Step 2: The paired transitions therefore coincide: 6 to 4 with 3 to 2; 6 to 2 with 3 to 1; and 4 to 2 with 2 to 1.

Step 3: The three distinct positive differences are

$$\frac{5E_0}{36}, \qquad \frac{8E_0}{9}, \qquad \frac{3E_0}{4}.$$

Why: Counting named transitions is different from counting observed photon-energy values. Nuclear-charge scaling exactly compensates for the doubled level labels in the specified pairs. The question restricts the transition set; it does not assert that an actual unrestricted He+ cascade lacks other lines.

Why not the others: (A) counts each species' coincident line separately. (B) merges only two pairs and misses the full level scaling. (D) incorrectly identifies two of the three unequal energy gaps as the same.

Key point: Compare energy differences after nuclear-charge scaling before counting distinct spectral lines.

Q21 / A Hydration stoichiometry and a later aliquot require separate mole accounts

Answer: (A) x=5 and 5.00×10^{-4} mol.

Given: Anhydrous residue is $0.500 - 0.180 = 0.320$ g; all of it enters the stated solution.

Formula/Principle: Infer hydrate stoichiometry from water-to-salt mole ratio, then use the aliquot fraction of a homogeneous solution.

Step 1:

$$n_{\text{salt}} = 0.320/160 = 0.00200 \text{ mol}, \quad n_{\text{water}} = 0.180/18.0 = 0.0100 \text{ mol}.$$

Step 2:

$$x = n_{\text{water}}/n_{\text{salt}} = 5.$$

Step 3:

$$n_{25} = 0.00200(25.0/100.0) = 5.00 \times 10^{-4} \text{ mol}.$$

Why: The starting mass includes water, so it cannot be divided directly by the anhydrous molar mass to count salt units. Heating separates the two mass contributions under the explicit no-decomposition assumption. Dilution changes concentration but conserves the salt amount, and a quarter-volume aliquot contains a quarter of those units.

Why not the others: (B) reports the whole solution's amount for the aliquot. (C) uses the wrong water-to-salt mole ratio despite the correct aliquot amount. (D) miscounts both hydrate water and the sampled fraction.

Key point: Separate crystallisation water from anhydrous formula units before handling solution aliquots.

Q22 / D Two eluents change an Rf value without proving chemical reaction or yield

Answer: (D) Rf increases from 0.20 to 0.60 without proving conversion.

Given: Distances are measured from each plate's own baseline to its spot and solvent front.

Formula/Principle: $R_f = d_{\text{spot}}/d_{\text{front}}$ for the specified stationary and mobile phases.

Step 1:

$$R_{f,X} = 2.0/10.0 = 0.20.$$

Step 2:

$$R_{f,Y} = 4.8/8.0 = 0.60.$$

Why: Migration reflects the compound's relative interactions with the stationary phase and the particular eluent. Changing the mobile phase can therefore change Rf for the same compound. Neither spot travel nor its ratio to the solvent front is a chemical yield measurement, and the data alone do not demonstrate a new covalent species.

Why not the others: (A) miscalculates the second ratio and gives it a reaction-yield meaning. (B) uses a complementary or incorrect distance for X and invents decomposition. (C) ignores dependence on the chromatographic conditions.

Key point: A chromatographic migration ratio is condition-dependent evidence, not an intrinsic reaction-progress meter.

Q23 / C The same internal alkyne follows stereoselective reduction or carbonyl-forming hydration

Answer: (C) cis-But-2-ene and butan-2-one.

Why: Controlled Lindlar hydrogenation adds hydrogen to an alkyne while stopping at the alkene, with the cis product from this internal alkyne. Hydration first gives an enol that converts to the more stable carbonyl form, here butan-2-one. But-2-yne is symmetric, so the hydration orientations do not create different constitutional ketones. Both routes retain all four carbons but lead to different functional groups.

Why not the others: (A) assigns the trans-selective outcome of a different reduction route to Lindlar conditions. (B) changes an internal-alkyne hydration product into an aldehyde. (D) over-reduces the first aliquot and treats the hydration enol route as ordinary ketone reduction.

Key point: Distinguish selective alkyne reduction from hydration followed by enol-carbonyl conversion.

Q24 / B Geminal and vicinal starting connectivity converge after two valid elimination stages

Answer: (B) P geminal, Q vicinal, and ethyne.

Why: In P both halogens are attached to one carbon, whereas in Q they are on adjacent carbons. Those arrangements define geminal and vicinal dihalides, respectively. Removing two HBr units under the stated sufficiently strong conditions creates a carbon-carbon triple bond in either two-carbon substrate. The explicit second-stage condition avoids assuming an ordinary mild base always completes elimination from a vinyl halide.

Why not the others: (A) reverses the two classifications and removes only one degree of unsaturation. (C) keeps the classifications but replaces elimination by reduction to a saturated hydrocarbon. (D) reverses the classifications and stops at the first elimination product.

Key point: Classify halogen positions separately from how far a staged elimination proceeds.

Q25 / D A phenol control prevents misclassifying every negative Lucas test as a primary alcohol

Answer: (D) The test response must be interpreted within the functional-group class.

Why: The tertiary aliphatic alcohol readily undergoes the substitution that produces turbidity under Lucas conditions. In phenol the OH is attached directly to an aromatic carbon; its bonding and substitution behaviour differ from the aliphatic carbinol systems used for the usual primary-secondary-tertiary Lucas comparison. Failure to give that response therefore does not assign phenol to the primary-alcohol category. A negative result narrows an interpretation only after the sample's functional-group context is checked.

Why not the others: (A) contradicts the known OH in phenol. (B) invents a tertiary alkyl carbocation at an aromatic carbon and an unmeasured optical concealment. (C) applies the aliphatic carbon-substitution classification to the wrong carbon type.

Key point: Do not extend a diagnostic-test classification beyond the functional groups for which it is defined.

Q26 / A Selective peptide hydrolysis preserves a disulfide-linked pair of residues as one molecule

Answer: (A) Three water molecules and three separate product molecules.

Given: Four residues form one linear peptide backbone; a side-chain S-S connection joins the two terminal cysteine residues.

Formula/Principle: Count peptide bonds for hydrolysis, then count connected covalent products after those bonds alone are removed.

Step 1: A linear four-residue backbone has

$$N_{\text{peptide}} = 4 - 1 = 3$$

peptide bonds, consuming three water molecules.

Step 2: Glycine and alanine each become one separate molecule. The two cysteine-derived residues remain connected to one another through S-S, giving one cystine molecule.

Step 3:

$$N_{\text{products}} = 1 + 1 + 1 = 3.$$

Why: The side-chain crosslink does not add a peptide bond, but it does keep two residues in one covalent product after the backbone is hydrolysed. Counting residues alone would correctly describe four amino-acid units but incorrectly describe four separate molecules. The treatment's bond selectivity is essential to this distinction.

Why not the others: (B) silently reduces or cleaves the retained disulfide bond. (C) counts the S-S link as another water-consuming peptide hydrolysis. (D) undercounts the linear backbone's three peptide connections.

Key point: Distinguish bond count from connected-product count when a peptide has a nonpeptide crosslink.

Q27 / C Equal-pH buffers respond in different equilibrium regimes to the same acid dose

Answer: (C) 2.00 and $9 + \log(2/3)$.

Given: Equal initial ratios give equal initial pH, but X contains one-tenth the buffer amount.

Formula/Principle: Neutralise B first; only a solution retaining both B and BH+ can use the buffer relation $\text{pH} = 14 - \text{p}K_b + \log(n_B/n_{BH^+})$.

Step 1: In X all 0.0100 mol B is protonated. Excess strong acid is

$$0.0200 - 0.0100 = 0.0100 \text{ mol}, \quad [H^+] \simeq 0.0100 \text{ M}, \quad \text{pH}_X = 2.00.$$

Step 2: In Y,

$$n_B = 0.100 - 0.0200 = 0.0800, \quad n_{BH^+} = 0.100 + 0.0200 = 0.120 \text{ mol}.$$

Step 3:

$$\text{pH}_Y = 9.00 + \log(0.0800/0.120) = 9 + \log(2/3) \approx 8.82.$$

Why: Initial pH constrains a ratio, not the total amount available to absorb acid. The same dose exhausts X's base and leaves a strong-acid excess, while Y remains a buffer. BH+ has Ka=10^-9, so its additional dissociation in X is negligible beside the supplied 0.0100 M strong acid.

Why not the others: (A) inverts Y's surviving base-to-acid ratio. (B) treats equal initial pH as unlimited resistance to added acid. (D) uses all added HCl rather than the excess after neutralisation in X.

Key point: Buffer capacity and buffer pH are different properties; change the model when one partner is exhausted.

Q28 / D Measured liquid and vapour compositions determine a partial-vaporisation amount by conservation

Answer: (D) One-third of the moles are vapour, which is enriched in A.

Given: Feed fraction z=0.40, liquid fraction x=0.30, vapour fraction y=0.60.

Formula/Principle: Conserve total moles and the moles of A across the two phases.

Step 1: Let v be the fraction of feed moles in vapour, leaving 1-v in liquid.

Step 2: A balance gives

$$0.40 = 0.30(1 - v) + 0.60v \Rightarrow 0.10 = 0.30v \Rightarrow v = 1/3.$$

Step 3: Since $0.60 > 0.40 > 0.30$, vapour is enriched and liquid depleted in A relative to the original feed.

Why: A phase composition is not the fraction of feed that entered that phase. The two composition measurements must be combined with conservation of A to infer phase amounts. The calculation does not assume equal liquid and vapour quantities or identify either phase composition with the original mixture.

Why not the others: (A) reports the fraction remaining liquid as the vapour fraction. (B) averages the two compositions as if phase amounts had to be equal. (C) obtains the fraction but reverses the enrichment comparison.

Key point: Use component balance to separate phase amounts from phase compositions.

Q29 / A Free-ion complexation and reaction scaling separate intensive voltage from extensive Gibbs change

Answer: (A) 0.1816 V and Gibbs-change ratio 2.

Given: Pure solid activities are one; the original electron count is two and Nernst activities are those of participating free ions.

Formula/Principle: Scaling a reaction scales both its electron count and the logarithm of its quotient; cell potential is intensive, while $\Delta G = -nFE$ follows the written reaction amount.

Step 1:

$$Q = \frac{a(\mathsf{A}^{2+})}{a(\mathsf{B}^{2+})} = 10^{-2}/10^{-6} = 10^{4}.$$

Step 2: For the original equation,

$$E = 0.3000 - (0.0592/2)(4) = 0.1816 \text{ V}.$$

Step 3: The doubled equation has $n' = 4$ and $Q' = Q^2 = 10^8$, giving

$$E' = 0.3000 - (0.0592/4)(8) = 0.1816 \text{ V}.$$

Step 4:

$$\frac{\Delta G'}{\Delta G} = \frac{-4FE'}{-2FE} = 2.$$

Why: Complexed B is not interchangeable with the explicitly participating free ion in the quotient. Rewriting a balanced equation does not alter the physical voltage, because the simultaneous quotient and electron-count changes cancel. It does double the reaction's specified amount and therefore its Gibbs-energy change.

Why not the others: (B) doubles n without also squaring Q. (C) correctly keeps the voltage but wrongly treats the extensive Gibbs change as invariant. (D) doubles the logarithm while retaining the original electron count.

Key point: Use free participating activities and scale Q, n and reaction energy consistently.

Q30 / B A phase-change Hess step precedes a phase-consistent gas-mole energy correction

Answer: (B) -242.0 and -240.75 kJ/mol.

Given: The requested water phase is gas, not the liquid in the supplied formation reaction.

Formula/Principle: Add the phase-change enthalpy by Hess's law, then use $\Delta H = \Delta U + \Delta n_g RT$.

Step 1:

$$\Delta H_g = -286.0 + 44.0 = -242.0 \text{ kJ/mol}.$$

Step 2: Gas mole change for $\mathsf{H_2} + \frac{1}{2}\,\mathsf{O_2} \longrightarrow \mathsf{H_2O(g)}$ is

$$\Delta n_g = 1 - (1 + 1/2) = -1/2.$$

Step 3:

$$\Delta U_g = \Delta H_g - \Delta n_g RT = -242.0 - (-1/2)(2.50) = -240.75 \text{ kJ/mol}.$$

Why: A phase label affects both the reaction enthalpy and the gas-mole work correction. Adding vaporisation changes the product to the requested state before the internal-energy conversion is made. Using the liquid reaction's gas-mole change after adding the vaporisation enthalpy would combine two inconsistent reactions.

Why not the others: (A) solves the liquid-water reaction instead. (C) reverses the sign of the gas-mole correction. (D) keeps the liquid reaction's -1.5 gas-mole change for a gaseous-water product.

Key point: Fix the full reaction, including phases, before converting enthalpy to internal energy.

Mathematics

Q31 / A Unique parameter from cyclic commutation and nonsymmetric determinant

Answer: (A) 676.

Given: $AB_t = B_tA$.

Formula/Principle: Left multiplication by A cycles rows; right multiplication cycles columns. Their equality requires a circulant pattern.

Step 1: The off-diagonal entries already follow that pattern. The diagonal entries must satisfy

$$t^2 = 2t + 3 = 15 - 2t.$$

Step 2: Equality of the last two gives t=3; this also satisfies the first. The root -1 of $t^2 = 2t + 3$ fails the third equality. Thus $B = \begin{pmatrix} 9 & 3 & 1 \\ 1 & 9 & 3 \\ 3 & 1 & 9 \end{pmatrix}$.

Step 3:

$$\det B = 9(81 - 3) - 3(9 - 9) + (1 - 27) = 676.$$

Worked derivation:

$$t = 3 \;\Rightarrow\; \det B = 702 + 0 - 26 = 676.$$

Why: Commutation constrains several entries simultaneously; satisfying only one diagonal equality is insufficient.

Why not the others: (B) It uses the wrong sign for a cyclic determinant term. (C) Commuting matrices need not be singular; this determinant is nonzero. (D) It subtracts the cubic off-diagonal term incorrectly.

Key point: Solve all commutation constraints before evaluating the determinant.

Q32 / B Three overlapping forbidden-person position sets by inclusion-exclusion

Answer: (B) 28.

Given: Three forbidden events concern the positions of A, B and C.

Formula/Principle: Inclusion–exclusion must count overlapping forbidden placements without assigning two people the same position.

Step 1: Each event has $2 \cdot 4! = 48$ arrangements. The pair intersections have 3,4,3 legal ordered position pairs, giving $18, 24, 18$ arrangements.

Step 2: For all three events there are four distinct-position triples, each allowing $2!$ arrangements for D,E: total 8.

Step 3: Subtract the union from 5!.

Worked derivation:

$$N = 120 - 3(48) + (18 + 24 + 18) - 8 = 28.$$

Why: The shared forbidden positions make the pair intersections unequal; independent multiplication of allowed counts would permit collisions.

Why not the others: (A) It omits the triple-overlap subtraction. (C) It does not retain the larger A–C intersection. (D) It subtracts the triple intersection twice.

Key point: Overlapping forbidden positions require intersection-specific counts.

Q33 / C Equal-image equivalence classes separated from codomain labels

Answer: (C) 60 relations and 24 maps per relation.

Given: The three classes have unequal sizes and must correspond to three distinct images.

Formula/Principle: Equal-image classes are the nonempty fibres of f.

Step 1: Choose the three-element class and then the two-element class; the singleton is forced. Their different sizes distinguish them.

$$N_R = \binom{6}{3}\binom{3}{2} = 20 \cdot 3 = 60.$$

Step 2: Assign three different elements of B to these fixed classes.

$$N_f = 4 \cdot 3 \cdot 2 = 24.$$

Worked derivation:

$$(N_R, N_f) = (60, 24).$$

Why: Reusing an image for two classes merges them and changes the induced relation. Dividing the class selection by 3! would also be wrong because sizes identify classes.

Why not the others: (A) 64 allows image collisions. (B) 10 divides the class count by an unjustified 3!. (D) Four selects only one image rather than three distinct images.

Key point: A relation fixes fibres, while the codomain supplies distinct fibre labels.

Q34 / D Repeated complex root imposes simultaneous real constraints

Answer: (D) 11.

Given: A monic quadratic has a repeated root.

Formula/Principle: Its discriminant is zero, equivalently its constant term is the square of half its negative linear coefficient.

Step 1:

$$(a+i)^2 = b + 6i \;\Rightarrow\; (a^2 - 1) + 2ai = b + 6i.$$

Step 2: Equate imaginary parts: 2a=6 gives a=3. Equate real parts: b=9-1=8.

Worked derivation:

$$a + b = 3 + 8 = 11; \qquad P(z) = (z - (3+i))^2.$$

Why: A complex equality imposes both real and imaginary constraints; one real discriminant condition is insufficient.

Why not the others: (A) Eight is b alone. (B) It drops the negative sign in i squared when combining coefficients. (C) It uses b=a squared instead of a squared minus one.

Key point: Apply the repeated-root condition as a complex equality.

Q35 / A Rational primitive with continuous arctangent branch and endpoint cancellation

Answer: (A) π.

Given: The denominator is positive for every real x, so no pole splits the interval.

Formula/Principle: Complete the square and split the numerator into a derivative-of-denominator part and a residual part.

Step 1: Set u=x-1. Then x+1=u+2 and the endpoints become -1 and 1.

Step 2:

$$F(2) = \int_{-1}^{1} \frac{u+2}{1+u^2}\,du = \left[\tfrac{1}{2}\ln(1+u^2) + 2\arctan u\right]_{-1}^{1}.$$

Worked derivation:

$$F(2) = 0 + 2(\pi/4 + \pi/4) = \pi.$$

Why: The logarithmic terms cancel at symmetric endpoints. The ordinary continuous principal arctangent gives a valid antiderivative across the whole interval.

Why not the others: (B) It loses the residual numerator factor 2. (C) It retains a logarithmic contribution that cancels. (D) It doubles the already evaluated arctangent difference.

Key point: Completing the square exposes both logarithmic and arctangent contributions.

Q36 / B Spatial section ratio combined with two coordinate-plane intercepts

Answer: (B) B=(3,0,6) and AB=3√14.

Given: A=(0,a,c), B=(b,0,d), and P=(A+2B)/3.

Formula/Principle: Internal section formula and a direction perpendicular to the plane normal.

Step 1: Coordinates give $b = 3, a = 9, c + 2d = 12$.

Step 2: A direction vector is $(3, -9, d - c)$. Parallelism gives

$$3 - 9 + d - c = 0 \;\Rightarrow\; d - c = 6.$$

Together with c+2d=12 this gives c=0,d=6.

Step 3:

$$AB = \sqrt{3^2 + (-9)^2 + 6^2} = \sqrt{126} = 3\sqrt{14}.$$

Worked derivation:

$$A = (0, 9, 0), \quad B = (3, 0, 6), \quad (A + 2B)/3 = (2, 3, 4).$$

Why: The section ratio fixes two intercept coordinates but the plane condition is needed to determine both heights.

Why not the others: (A) It reverses the section weights. (C) Its direction is not parallel to the plane. (D) It removes an unjustified factor 3 from the length.

Key point: Combine coordinate-plane intercepts, section ratio and directional constraints.

Q37 / A Separate a regular vertical tangent from two crossing regular parameter branches

Answer: (A) The origin is regular with a vertical tangent; the self-intersection has perpendicular tangents.

Given: The entire real parameter domain is included.

Formula/Principle: Distinct parameters at one point can produce different tangent branches; regularity requires a nonzero velocity vector.

Step 1: At the origin t=0, $x\prime = 2t = 0$ and $y\prime = 3t^2 - 1 = -1$. The velocity is nonzero and the tangent is x=0.

Step 2: The point (1,0) has two preimages, t=1 and t=-1. Both are regular, with slopes

$$m = \frac{3t^2 - 1}{2t} = 1 \quad \text{and} \quad -1.$$

Worked derivation:

$$m_1 m_2 = -1 \;\Rightarrow\; \theta = \pi/2.$$

Why: A vertical tangent at one regular parameter is different from a geometric self-intersection with two regular parameter branches.

Why not the others: (B) The velocity at zero is (0,-1), not zero. (C) It exchanges horizontal and vertical tangency and miscounts the crossing angle. (D) The two slopes at (1,0) have opposite signs and are distinct.

Key point: Check both parameter preimages and their velocities before classifying a parametric curve.

Q38 / B Outlier reflection changes location while preserving two dispersion measures

Answer: (B) Both stated dispersion measures are unchanged.

Given: The second data set is the reflection of the first about zero.

Formula/Principle: Variance averages squared deviations; mean absolute deviation averages absolute deviations.

Step 1: The means are 1 and -1. For either set the deviation magnitudes are 1,1,1,3.

Step 2:

$$\sigma^2 = (1+1+1+9)/4 = 3, \qquad MAD = (1+1+1+3)/4 = 3/2.$$

Worked derivation: Reflection changes each signed deviation to its negative and preserves both its square and its magnitude.

Why: A sign change of the outlier changes the mean but does not necessarily change dispersion. Both definitions must be checked rather than inferred from the mean.

Why not the others: (A) It confuses location change with dispersion change. (C) The absolute-deviation sum remains 6. (D) It uses an incorrect mean or divisor.

Key point: Reflection preserves dispersion about the reflected mean.

Q39 / C Cancel constant terms and use the second-order limit to recover a parameter difference

Answer: (C) -12.

Given: A finite two-sided limit after division by x².

Formula/Principle: Constant and linear terms must cancel before the quadratic contribution can remain finite.

Step 1: The constant term is a+b; finiteness forces b=-a. The linear term of sin x cancels the displayed -x.

Step 2:

$$\frac{\sin x - x}{x^2} + a\frac{\cos x - 1}{x^2} \longrightarrow 0 - a/2.$$

The value 3 gives a=-6 and b=6.

Worked derivation:

$$a - b = -6 - 6 = -12.$$

Why: The standard trigonometric limits evaluate the remaining second-order term only after the lower-order terms vanish.

Why not the others: (A) It reports a alone. (B) It reports b alone. (D) It reports the limit rather than the requested parameter difference.

Key point: Enforce finiteness before evaluating the surviving order.

Q40 / D Unequal independent match probabilities and directed tournament cycles

Answer: (D) 17/24.

Given: Three independent pairwise matches with unequal probabilities.

Formula/Principle: A three-player tournament is either transitive or one of two directed cycles.

Step 1: The cycle A beats B beats C beats A has probability $(1/2)(2/3)(3/4) = 1/4$.

Step 2: The reverse cycle has probability $(1/2)(1/3)(1/4) = 1/24$. Therefore the cyclic probability is 7/24.

Worked derivation:

$$P(\text{transitive}) = 1 - 1/4 - 1/24 = 17/24.$$

Why: Both cycle orientations must be included before taking the complement. Unequal match probabilities prevent treating all eight outcomes as equally likely.

Why not the others: (A) One third does not equal the sum or complement of the two cycle probabilities. (B) It counts only the more likely cycle. (C) It excludes only that cycle.

Key point: Enumerate both directed cycles and weight each by its own probabilities.

Q41 / B Inverse-tangent sum after a principal-value boundary crossing

Answer: (B) Add π to the principal arctangent.

Given: Both original arctangents lie between zero and π/2.

Formula/Principle: The tangent addition formula determines an angle only modulo π.

Step 1:

$$\tan S = \frac{2x + x}{1 - 2x^2} = \frac{3x}{1 - 2x^2}.$$

For x>1/√2 the denominator is negative and numerator positive.

Step 2: S lies in (π/2,π), whereas the principal arctangent of this negative quotient lies in (-π/2,0).

Worked derivation:

$$S = \pi + \arctan \frac{3x}{1 - 2x^2}.$$

Why: Range information selects the correct representative of the tangent angle.

Why not the others: (A) It yields a negative angle. (C) It moves farther outside the required range. (D) It adds π/2, which changes the tangent rather than preserving it.

Key point: Recover a principal-value sum using both its tangent and its quadrant.

Q42 / D Linear differential equation with x dependent on y

Answer: (D) 13/6.

Given: x is the dependent variable and y is positive.

Formula/Principle: This is a first-order linear equation with integrating factor $e^{\int dy/y} = y$.

Step 1: Multiply by y:

$$y\frac{dx}{dy} + x = y^2 \;\Rightarrow\; \frac{d(yx)}{dy} = y^2.$$

Step 2: Integrate: $yx = y^3/3 + C$. The condition gives $2 = 1/3 + C$, so C=5/3.

Worked derivation:

$$x(2) = 4/3 + 5/6 = 13/6.$$

Why: Treating y as the independent variable exposes a linear equation; simply replacing dx/dy by dy/dx would change it.

Why not the others: (A) It does not satisfy the initial-value solution. (B) It drops the integration constant. (C) It replaces the integrating factor y by a constant.

Key point: Identify the dependent variable before applying a linear-equation formula.

Q43 / D Coefficient-exchanged distinct quadratics and a repeated common root

Answer: (D) p+q=-1.

Given: P and Q are distinct, so p≠q.

Formula/Principle: A common root must also annihilate their difference; a monic quadratic with a repeated root r is (x-r)².

Step 1:

$$P(x) - Q(x) = (p - q)(x - 1).$$

Since p≠q, the common root is 1.

Step 2: Repetition in P forces $P = (x-1)^2 = x^2 - 2x + 1$, hence p=-2 and q=1.

Worked derivation:

$$p + q = -1, \qquad Q = x^2 + x - 2 = (x - 1)(x + 2).$$

There is exactly one distinct common root, as required.

Why: Subtraction identifies the only possible common root before multiplicity fixes the two coefficients.

Why not the others: (A) It reports q alone. (B) It reports p-q rather than the sum. (C) It assumes equal opposite coefficients, incompatible with the repeated root.

Key point: Verify the distinct-polynomial condition before cancelling a parameter factor.

Q44 / A Admissible hyperbola-circle chords after excluding axis-parallel point pairs

Answer: (A) Two lines, each with chord length 6.

Given: Both endpoints satisfy both conic equations; horizontal and vertical joining lines are excluded.

Formula/Principle: Solve the simultaneous equations for squared coordinates, then classify the possible point pairs.

Step 1:

$$x^2 - 4y^2 = 4, \quad x^2 + y^2 = 9 \;\Rightarrow\; y^2 = 1, \quad x^2 = 8.$$

The four intersection points are $(\pm 2\sqrt{2}, \pm 1)$.

Step 2: Of their six joining pairs, two give horizontal lines and two vertical lines. The other two join opposite corners, hence antipodal circle points.

Worked derivation:

$$N = 6 - 2 - 2 = 2, \quad L = \sqrt{(4\sqrt{2})^2 + 2^2} = 6.$$

Why: Excluding axis-parallel pairs selects only the two diameters; both endpoints must still be actual hyperbola-circle intersections.

Why not the others: (B) It retains forbidden axis-parallel lines. (C) Three is the circle radius, not the diameter. (D) It uses only horizontal separation and omits the vertical difference.

Key point: Use conic intersections to enumerate admissible chords before measuring them.

Q45 / C Projection and signed area determine perpendicular components

Answer: (C) √65.

Given: The perpendicular basis a,J(a) has both squared lengths 5 and positive determinant 5.

Formula/Principle: Write v=αa+βJ(a); orthogonality separates projection from oriented area.

Step 1: $v \cdot a = 5\alpha = 10$ gives α=2. The signed determinant gives $5\beta = 15$, so β=3.

Step 2:

$$v = 2(2, 1) + 3(-1, 2) = (1, 8).$$

Worked derivation:

$$|v| = \sqrt{1 + 64} = \sqrt{65} = \sqrt{5(2^2 + 3^2)}.$$

Why: The determinant fixes the signed perpendicular component, not the parallel projection.

Why not the others: (A) It ignores the unequal component coefficients. (B) It treats the basis vectors as unit vectors. (D) Its squared length 85 contradicts the component sum 5(4+9)=65.

Key point: Normalise a perpendicular basis when converting coefficients to length.

Physics

Q46 / A Infer three drag exponents while retaining an undetermined coefficient

Answer: (A) The exponents are 1,2,2, but K is undetermined.

Given: Dimensions are [F]=MLT^{-2}, [ρ]=ML^{-3} and [v]=LT^{-1}.

Formula/Principle: Independent base-dimension exponents must match.

Step 1: The right side has dimensions $M^a L^{-3a+b+c} T^{-b}$. Thus a=1 and b=2.

Step 2: Length matching gives -3+2+c=1, hence c=2.

Worked derivation:

$$[\rho v^2 L^2] = MLT^{-2}.$$

Why: Every dimensionless multiplier has the same dimensions, so dimensional consistency cannot determine its numerical value or establish the law experimentally.

Why not the others: (B) b=1 gives the wrong time dimension. (C) c=1 leaves length exponent zero. (D) No dimensional equation determines K.

Key point: Dimensional analysis constrains powers, not dimensionless constants.

Q47 / B A negative trial normal invalidates common block acceleration

Answer: (B) The blocks separate with accelerations -3 and +6 m s^{-2}.

Given: A repulsive normal contact is the only possible interaction between blocks.

Formula/Principle: A common-acceleration assumption must produce a nonnegative normal force.

Step 1: If they stayed together, a=(12-3)/3=3 m s^{-2}. For A this would require

$$-3 - N = 1(3) \;\Rightarrow\; N = -6 \text{ N}.$$

Step 2: Negative N would be tension, impossible for simple contact. Therefore N=0.

Worked derivation:

$$a_A = -3/1 = -3, \qquad a_B = 12/2 = 6 \text{ m s}^{-2}.$$

Why: The right block accelerates away faster, so the contact opens rather than pulling the left block along.

Why not the others: (A) It retains an impossible tensile normal. (C) It reverses the applied force directions. (D) Neither block is in force balance.

Key point: Check unilateral contact constraints before combining masses.

Q48 / C Parallel support redistribution after an explicitly supplied plastic plateau

Answer: (C) Forces are 90 N and 150 N, and extension ratio is 6/5.

Given: Equal extensions follow from the guided platform; post-yield behaviour is explicitly stipulated.

Formula/Principle: Before yield, Fi=kiδ; afterwards rod 2's force is fixed while force balance still holds.

Step 1: Rod 2 first yields at $\delta_y = 150/(2k) = 75/k$. Rod 1 then carries 75 N, below its limit; total load is 225 N.

Step 2: At 240 N rod 2 stays at 150 N. Rod 1 must carry 240-150=90 N, still below 100 N.

Worked derivation:

$$\delta = 90/k, \qquad \delta/\delta_y = (90/k)/(75/k) = 6/5.$$

Why: The initial 1:2 force ratio applies only while both rods are elastic. The supplied plastic plateau redistributes additional load to rod 1.

Why not the others: (A) It extrapolates rod 2 beyond its plateau. (B) It puts rod 1 at its limit before the load requires that. (D) Equal extension does not imply equal force.

Key point: Rebuild the force-sharing model after a constitutive regime changes.

Q49 / D Three cooling readings determine unknown ambient and a fourth temperature

Answer: (D) 145/3 degrees C.

Given: Equal time intervals and a fixed ambient temperature Ta.

Formula/Principle: Excess temperature decays by the same multiplicative factor over each interval.

Step 1:

$$(65 - T_a)^2 = (80 - T_a)(55 - T_a).$$

Expanding gives 4225-130Ta=4400-135Ta, hence Ta=35 degrees C.

Step 2: The interval factor is $(65 - 35)/(80 - 35) = 2/3$.

Worked derivation:

$$T_4 = 35 + (55 - 35)(2/3) = 145/3\ ^\circ\text{C}.$$

Why: Temperature differences, not absolute Celsius readings, decay geometrically relative to the ambient.

Why not the others: (A) It subtracts the previous 10-degree fall unchanged. (B) It assumes an arbitrary shrinking arithmetic decrement. (C) It repeats the first 15-degree drop.

Key point: Infer the ambient before extrapolating exponential cooling.

Q50 / A Energy-weighted mixing before comparing molecular rms speeds

Answer: (A) √(5/6).

Given: Same gas, no external work and no heat exchange.

Formula/Principle: Internal energy is 3nRT/2 and rms speed is proportional to √T at fixed molecular mass.

Step 1:

$$T_f = \frac{1(100) + 3(300)}{4} = 250\ \text{K}.$$

Step 2:

$$v_f/v_{hot} = \sqrt{250/300} = \sqrt{5/6}.$$

Worked derivation: The molecular mass and factor 3R cancel only after the common final temperature has been found.

Why: Conservation of energy requires mole-weighted temperature, not an unweighted average of two samples.

Why not the others: (B) It uses the unweighted 200 K average. (C) It makes speed linear in temperature. (D) It uses the mole-number ratio as a speed ratio.

Key point: Mix energies first, then convert temperature to molecular speed.

Q51 / B Quarter-period force removal and the new free-oscillation amplitude

Answer: (B) √2 F/k.

Given: Turning a force off changes acceleration, not instantaneous position or velocity.

Formula/Principle: During forcing the shifted equilibrium is d=F/k; afterwards free amplitude obeys $A^2=x^2+(v/\omega)^2$.

Step 1: With the stated initial conditions,

$$x = d(1 - \cos\omega t), \quad v = d\omega \sin\omega t.$$

Step 2: At the quarter-period switch, x=d and v=dω.

Worked derivation:

$$A = \sqrt{d^2 + (d\omega/\omega)^2} = \sqrt{2}\,F/k.$$

Why: The mass has both displacement and kinetic energy when the force is removed; the shifted equilibrium is no longer the centre of motion.

Why not the others: (A) It counts displacement but omits velocity. (C) It uses the half-period forced excursion instead of the given switch time. (D) It divides rather than combines the two equal energy contributions.

Key point: Match both position and velocity across a force switch.

Q52 / C Node-spacing frequency with a free-end harmonic constraint

Answer: (C) 100 Hz, fifth harmonic.

Given: Adjacent nodes are separated by 0.40 m.

Formula/Principle: Node spacing is λ/2; fixed–free fundamental wavelength is 4L.

Step 1: $\lambda = 0.80$ m gives $f = v/\lambda = 80/0.80 = 100$ Hz.

Step 2: $f_1 = v/(4L) = 20$ Hz, so f/f1=5.

Worked derivation:

$$L = 5\lambda/4 = 1.00 \text{ m},$$

consistent with an antinode one quarter-wavelength beyond the last node.

Why: Fixed–free boundaries admit odd harmonics and differ from the two-fixed-end spectrum.

Why not the others: (A) It uses node spacing as wavelength. (B) It mislabels the harmonic relative to 20 Hz. (D) It conflicts with both observed spacing and speed.

Key point: Infer wavelength locally and verify it against both boundaries.

Q53 / D Ground-disconnect-deposit sequence separates cavity asymmetry from exterior symmetry

Answer: (D) The inner charge is -q, outer charge +2q, with spherical exterior field.

Given: Grounding occurs before disconnection and the extra deposition; the off-centre charge remains fixed.

Formula/Principle: A Gaussian surface inside conductor material encloses zero net charge. A grounded conductor with no external sources initially has zero exterior field.

Step 1: Grounding supplies -q to the inner wall and leaves zero outer charge. The conductor's charge at disconnection is therefore -q.

Step 2: Adding +2q changes total conductor charge to +q. The inner wall must still carry -q, so the outer wall carries +2q.

Step 3: The spherical outer surface is equipotential with no external sources, giving a uniform outer density and spherical exterior solution.

Worked derivation:

$$Q_{cond} = q, \quad Q_{outer} = q - (-q) = 2q, \quad E(r) = 2kq/r^2 \ (r > R).$$

Why: The cavity's asymmetric induced distribution is screened from the exterior; grounding fixes the initial charge budget rather than leaving the conductor neutral.

Why not the others: (A) It assumes the conductor was neutral after grounding. (B) It violates zero field within metal. (C) It carries cavity asymmetry into an exterior bounded by a spherical equipotential.

Key point: Track grounding and isolation in order before invoking exterior symmetry.

Q54 / A Nonohmic operating point from a quadratic characteristic and source load line

Answer: (A) (√13-1) A.

Given: The operating point lies on both the device curve and the source line.

Formula/Principle: The terminal voltage must be the same in the two relations.

Step 1: In the stated SI numerical units,

$$I^2 = 12 - 2I \;\Rightarrow\; I^2 + 2I - 12 = 0.$$

Step 2: The two roots are $-1 \pm \sqrt{13}$; only the positive one belongs to I≥0.

Worked derivation:

$$I = \sqrt{13} - 1 \approx 2.61 \text{ A}, \quad V = I^2 \approx 6.79 \text{ V}.$$

Why: The source's internal resistance lowers its terminal voltage as current increases. The device is nonohmic, so a fixed resistance cannot be assumed.

Why not the others: (B) At 3 A the device needs 9 V but the source supplies 6 V. (C) At 6 A the source terminal voltage is zero. (D) It changes the sign of the linear-root offset and fails the equation.

Key point: Solve the characteristic and load line simultaneously.

Q55 / B Transformer winding polarity in aiding and opposing series connections

Answer: (B) X gives 10 V and Y gives 2 V.

Given: Ideal secondary emfs are in phase at their dotted ends.

Formula/Principle: Voltage ratio equals turn ratio; series connections add or subtract signed emfs according to winding polarity.

Step 1:

$$V_{40} = 10(40/100) = 4 \text{ V}, \quad V_{60} = 10(60/100) = 6 \text{ V}.$$

Step 2: Undotted-to-dotted connection adds the emfs; dotted-to-dotted opposes them.

Worked derivation:

$$V_X = 4 + 6 = 10 \text{ V}, \quad V_Y = |6 - 4| = 2 \text{ V}.$$

Why: RMS magnitudes can be added directly here only after the relative phase implied by the dot connections is determined.

Why not the others: (A) It reverses aiding and opposing connections. (C) It ignores polarity in Y. (D) It reports individual winding voltages instead of free-end voltages.

Key point: Determine series polarity before combining transformer emfs.

Q56 / C Reactive component voltage exceeding supply under phasor cancellation

Answer: (C) 16 V; the reactive voltage phasors partially cancel.

Given: Net reactance is 8-4=4 ohm.

Formula/Principle: $Z = \sqrt{R^2 + (X_L - X_C)^2}$, with VR and net reactive voltage in quadrature.

Step 1: Z=√(9+16)=5 ohm, so I=10/5=2 A rms.

Step 2: VL=2×8=16 V, VC=8 V and VR=6 V.

Worked derivation:

$$V_s = \sqrt{V_R^2 + (V_L - V_C)^2} = \sqrt{36 + 64} = 10 \text{ V}.$$

Why: Individual reactive voltage magnitudes need not be bounded by their vector sum with the other voltages. Ideal L and C exchange energy without net average dissipation.

Why not the others: (A) It both miscomputes VL and uses scalar addition. (B) A large reactive voltage does not create energy. (D) It imposes a false component-voltage bound.

Key point: Apply phasor addition, not arithmetic addition of rms magnitudes.

Q57 / D Single-surface refraction changes image reality across the object focal position

Answer: (D) -180 cm and +180 cm.

Given: n1=1,n2=1.5,R=+20 cm; positive x follows incident light.

Formula/Principle: $n_2/v - n_1/u = (n_2 - n_1)/R$.

Step 1: For u=-30 cm,

$$1.5/v + 1/30 = 1/40 \;\Rightarrow\; 1.5/v = -1/120, \quad v = -180 \text{ cm}.$$

Step 2: For u=-60 cm,

$$1.5/v + 1/60 = 1/40 \;\Rightarrow\; 1.5/v = 1/120, \quad v = 180 \text{ cm}.$$

Worked derivation: The nearer object gives a virtual image on the incident side; the farther object gives a real image in glass.

Why: This is refraction at one surface, so the two refractive indices remain in the conjugate equation; the thin-lens formula is inappropriate.

Why not the others: (A) It loses the negative result for the nearer object. (B) It wrongly makes the farther image virtual. (C) It does not satisfy the spherical-surface relation for either object.

Key point: Keep signed object distance and curvature while distinguishing a surface from a lens.

Q58 / C Source displacement changes Young phase and reverses the zero-fringe shift

Answer: (C) The old centre becomes dark; the new zero-difference fringe is at -β/2.

Given: Source and screen path differences both contribute.

Formula/Principle: Upper-minus-lower path difference is approximately $-d(s/L + y/D)$.

Step 1: At y=0, $\Delta = -ds/L = -\lambda/2$, a destructive phase difference.

Step 2: Zero total difference requires $s/L + y/D = 0$, giving y=-Ds/L.

Worked derivation:

$$y = -D\lambda/(2d) = -\beta/2.$$

Why: Moving the source shortens its path to the upper slit; moving the observed bright fringe downward compensates that imbalance.

Why not the others: (A) A half-wavelength phase difference is dark and the shift sign is wrong. (B) It gets darkness but reverses compensation direction. (D) It uses a full fringe shift and incorrect centre intensity.

Key point: Include source-to-slit geometry before computing the screen path difference.

Q59 / A Acceleration followed by partial retardation before a wavelength ratio

Answer: (A) 2/√3.

Given: The second potential removes 100 eV from the 400 eV kinetic energy.

Formula/Principle: $\lambda = h/\sqrt{2mK}$; successive energy changes must be combined with their signs.

Step 1: $K_i = 400$ eV and $K_f = 400 - 100 = 300$ eV, positive as required for transmission.

Worked derivation:

$$\lambda_f/\lambda_i = \sqrt{K_i/K_f} = \sqrt{400/300} = 2/\sqrt{3}.$$

Why: Retardation lowers momentum, increasing wavelength. The final energy is not simply the energy corresponding to the retarding voltage.

Why not the others: (B) It treats 100 eV as the final energy and reverses the ratio. (C) It inverts the wavelength ratio. (D) It makes wavelength inversely proportional to energy instead of its square root.

Key point: Sum signed kinetic-energy changes before converting to matter wavelength.

Q60 / B Battery reversal selects unequal antiparallel diode paths

Answer: (B) 60 mA then 30 mA.

Given: An ideal forward diode is a short and a reverse diode an open circuit.

Formula/Principle: Select the conducting path for each polarity and apply Ohm's law only to that path.

Step 1: Left positive forward-biases the 100 ohm branch: $I_1 = 6/100 = 0.060$ A.

Step 2: Right positive forward-biases the opposite 200 ohm branch: $I_2 = 6/200 = 0.030$ A.

Worked derivation:

$$(|I_1|, |I_2|) = (60, 30) \text{ mA}.$$

Why: Reversing the battery swaps which branch is closed; both branches are never simultaneously conducting in this ideal DC model.

Why not the others: (A) It reverses the two bias conditions. (C) It adds currents from a blocked branch. (D) Each polarity has one forward path.

Key point: Establish diode state before reducing the circuit.
